

Fuzzy Control of Fuel Sloshing in a Spacecraft

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Abstract—This paper studies effect of fuzzy controller on fuel sloshing dynamics in orbital maneuvering of the spacecraft. For this aim, a mathematical model of coupled dynamics of spacecraft and fuel sloshing is developed. Then, a classic controller is applied on equations of coupled motion, and because this controller has no good responsibility, therefore a fuzzy controller is suggested in this case. Also, its shown fuzzy controller is more robust than classic controller. In both designed controller law, the controller inputs are three moments about center of mass of the spacecraft and two gimbal deflections angles.

Index Terms—fuzzy controller, fuel slosh, coupled dynamics

I. INTRODUCTION

The attitude control is essential part in every space mission, and it contains two important part, mathematical modeling and attitude control [2], [3]. Orbital maneuvering is one of the attractive space missions. The high percent of mass of the spacecraft that maneuvering in space is allocated to fuel. Therefore, sloshing some of this high percent of the mass can easily disturb the attitude of the spacecraft [1]. So, fuel sloshing control is a critical object in attitude control. Achievement to nonlinear controllers in attitude control field is difficult and usually they are high nonlinear. So, classic controllers are suggested in these cases, but in some of them classic controller has no good responsibility. Therefore, a fuzzy controller [6] will be suggested. Furthermore, a fuzzy controller is more robust than classic controllers when uncertainties are included. Many studies have been done on the interaction between attitude of the spacecraft dynamics and fuel slosh dynamics. Most of these studies used mechanical modeling [4], [5] to model fuel sloshing.

These previous works, used nonlinear controller to control fuel sloshing. In this paper a 3D multi-pendulum modeling is used to model fuel sloshing. Therefore, equations of motion are high nonlinear, and achievement to nonlinear controller is too difficult. So, a classic controller is designed to control attitude of the spacecraft and fuel sloshing, and because this controller has no good responsibility, a fuzzy controller is suggested to this aim, and its shown that fuzzy controller is more robust than classic controller when uncertainties are included. As seen in Figure 1 the control inputs are three

moments about center of mass of the spacecraft (M_x , M_y , M_z) and two gimbal deflection angles (δ_1 , δ_2).

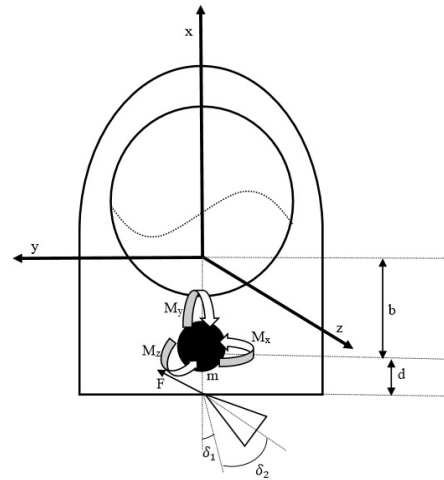


Fig. 1. A spacecraft with a partially filled fuel container

II. MODELING

In this section, equations of coupled dynamics of the spacecraft that modeled by a rigid body and maneuver in space, and fuel sloshing is derived. The spacecraft has a spherical fuel container, and fuel is modeled by two sections. One of these sections is un-turbulent fuel that modeled by a rigidly attached mass (m_0 , I_0), and the other section is sloshing mass that modeled by two pendulum masses (m_i , I_i , $i = 1, 2$).

Equations of the coupled dynamics of the spacecraft and fuel sloshing can derive by using Kirochoffs equations:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{V}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{V}} = \tau_t \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\Omega}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{\Omega}} + \bar{V} \times \frac{\partial L}{\partial \dot{V}} = \tau_r \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}} - \frac{\partial L}{\partial \zeta} + \frac{\partial R}{\partial \dot{\zeta}} = 0 \quad (3)$$

Where $V = \dot{r} + \bar{\Omega} \times r = [\dot{x} \ \dot{y} \ \dot{z}]^T + \bar{\Omega} \times [x \ y \ z]^T = [v_x \ v_y \ v_z]^T$ is denote vector of axial velocity of the rigid

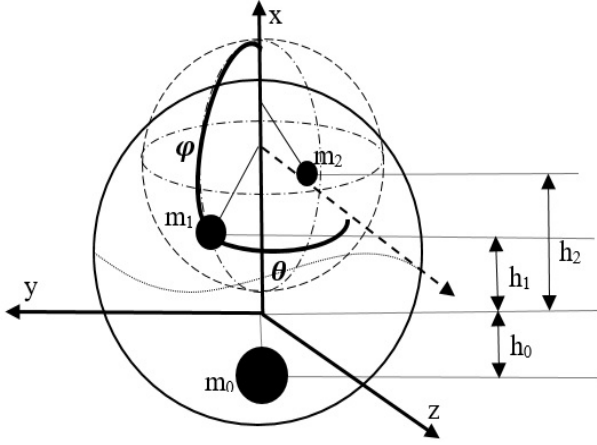


Fig. 2. A 3D multi-pendulum model of fuel sloshing

body, Also $\Omega = [\Omega_1 \ \Omega_2 \ \Omega_3]^T$ is denote vector of angular velocity of the rigid body, and as seen in Figure 2, $\zeta = [\varphi \ \theta]^T$ is vector of pendulums coordinates. For a given vector p , $\bar{p}$ is skew symmetric matrix, and τ_t, τ_r are vector of forces and moments that act on center of mass of the spacecraft. In above equations, R is Rayleigh dissipation function, and Also L is Lagrangian, and because we assume that the spacecraft maneuver in zero gravity environment, Lagrangian is equal to total kinetic energy. So, by applying (1-3), equations of the coupled dynamics can derive. These equations produce a dynamic system that it contain 8 equation of coupled motion of the spacecraft and fuel sloshing, but they are too large and high nonlinear. Therefore, we dont write them here.

III. CONTROLLER DESIGN

In this section, a feedback classic controller and a fuzzy controller are designed to control dynamic system that produced in previous section. We know that its not necessary to control the velocity vector along the spacecrafts longitudinal axis. So, we ignore first equation of the equations of dynamic system. Then, the other equations have to be written in reduced form. For this aim, by changing old control inputs (moments about rigid body and gimbal deflection angles) to new control inputs (u_{1-5}), these equations become to:

$$\dot{v}_y = u_1 - v_x \Omega_3 + v_z \Omega_1 \quad (4)$$

$$\dot{v}_z = u_2 - v_y \Omega_1 + v_x \Omega_3 \quad (5)$$

$$\dot{\Omega}_1 = u_3 \quad (6)$$

$$\dot{\Omega}_2 = u_4 \quad (7)$$

$$\dot{\Omega}_3 = u_5 \quad (8)$$

$$\begin{aligned} \ddot{\varphi}_i = & [A]v_y + [B]\dot{v}_y + [C]v_z + [D]\dot{v}_z + [E]\Omega_1 + [F]\dot{\Omega}_1 \\ & + [G]\Omega_2 + [H]\dot{\Omega}_2 + [K]\Omega_3 + [N]\dot{\Omega}_3 \end{aligned} \quad (9)$$

$$\begin{aligned} \ddot{\theta}_i = & [O]v_y + [P]\dot{v}_y + [Q]v_z + [S]\dot{v}_z + [T]\Omega_1 + [U]\dot{\Omega}_1 \\ & + [W]\dot{\Omega}_2 + [X]\Omega_3 + [Y]\dot{\Omega}_3 \end{aligned} \quad (10)$$

A. Classic Controller

By proportional (P) control theory for $u_{1,2}$ we can write:

$$u_1 = k_{p1}v_y \quad (11)$$

$$u_2 = k_{p2}v_z \quad (12)$$

Where, k_{p1} and k_{p2} are gains controller. Also, by proportional-derivative (PD) control theory for u_{3-5} can write:

$$u_3 = k_{p3}\vartheta_1 + k_{d3}\Omega_1 \quad (13)$$

$$u_4 = k_{p4}\vartheta_2 + k_{d4}\Omega_2 \quad (14)$$

$$u_5 = k_{p5}\vartheta_3 + k_{d5}\Omega_3 \quad (15)$$

Where, $\vartheta = [\vartheta_1 \ \vartheta_2 \ \vartheta_3]^T$ is vector of position angle of the spacecraft, and $k_{p3}, k_{d3}, k_{p4}, k_{d4}, k_{p5},$ and k_{d5} are gains controller.

B. Fuzzy Controller

The classic controller that designed in previous sub-section has no good responsibility. Therefore, a fuzzy controller is suggested. Consider the designed controllers in previous sub-section with these differences:

$$k_{pj} = k_{pj,min} + (k_{pj,max} - k_{pj,min})k'_{pj} \quad (16)$$

$$k_{dk} = k_{dk,min} + (k_{dk,max} - k_{dk,min})k'_{dk} \quad (17)$$

Where, k'_{pj} and k'_{dk} ($j = 1 - 5, k = 3 - 5$) are outputs of fuzzy mamdani logical. As seen in Figure 3 each output has two Gaussian membership function (B, S). Inputs of this fuzzy logical are $v_y, v_z, \vartheta_1, \dot{\vartheta}_1, \vartheta_2, \dot{\vartheta}_2, \vartheta_3,$ and $\dot{\vartheta}_3$, and as seen in Figure 4 each input has five tringle (NM, NS, ZO, PS, PM) and two trapeze (NB, PB) membership function.

The rules of this fuzzy logical for k'_{p3-5} and k'_{d3-5} are defined by using fuzzy tuning rules tables in [6]. Also, for defining rules between k'_{p1} and v_y , if v_y is NB or PB, then k'_{p1} is B, and if v_y is NM, NS, ZO, PS, or PM, then k'_{p1} is S. Defining rules between k'_{p2} and v_z is the same as k'_{p1} and v_y .

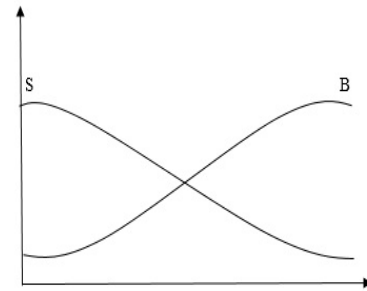


Fig. 3. Gaussian memberships function

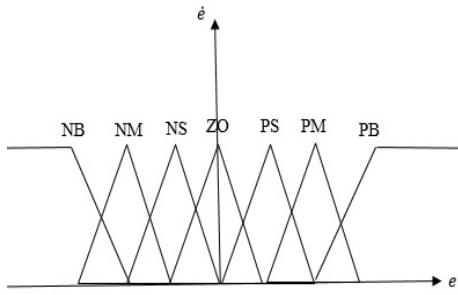


Fig. 4. Tringle and trapeze memberships function

IV. SIMULATION

In this section, a physical example is simulated to describe effectiveness of two control laws that designed in previous section. The control aims are control translation velocity vectors and attitude of the spacecraft, and attitude of the pendulums. The control gains are $k_{p1} = 4 \times 10^{-3}$, $k_{p2} = 4 \times 10^{-3}$, $k_{p3} = 8 \times 10^{-5}$, $k_{d3} = 5 \times 10^{-3}$, $k_{p4} = 8 \times 10^{-5}$, $k_{d4} = 5 \times 10^{-3}$, $k_{p5} = 8 \times 10^{-5}$, and $k_{d5} = 5 \times 10^{-3}$. Then by applying control laws on reduced equations of motion, we can see time responses of control effect on each variable. As seen in Figure 5 and Figure 6, the transverse velocity and attitude angles of the spacecraft after elapse 1800 second tend to zero, and fuzzy controller has better responsibility than classic controller. Also, as seen in Figure 7 and Figure 8 the position angle of the pendulums about x-axis tend to zero, and position angle of the pendulums about z-axis become stable, and fuzzy controller has better responsibility than classic controller. Time responses of the controller inputs is shown in Figure 9 and Figure 10.

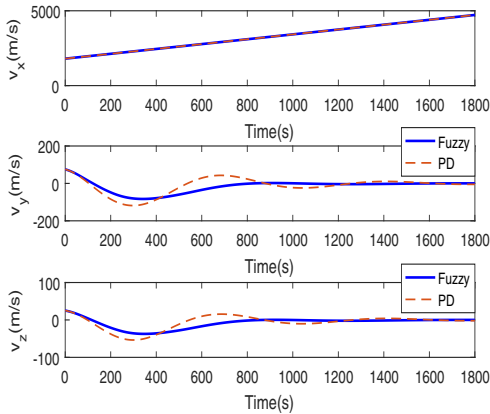


Fig. 5. Time responses for velocity vectors

Now for check robustness of fuzzy controller we decrease 40 percent of moment of inertia of the spacecraft to include uncertainties, and its observed classic controller not able to stabilize attitude of the spacecraft and fuel sloshing after elapse 1800 second, and it takes twice as much time as without uncertainties condition. Whereas, fuzzy controller

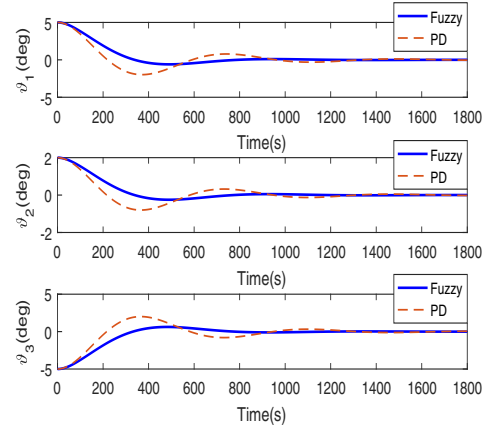


Fig. 6. Time responses for attitude angles of the spacecraft

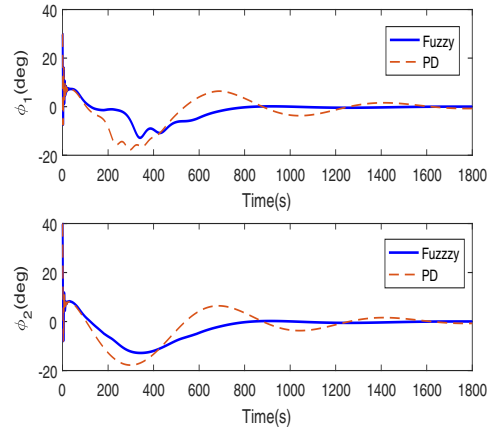


Fig. 7. Time responses for attitude angle about x-axis of the pendulums

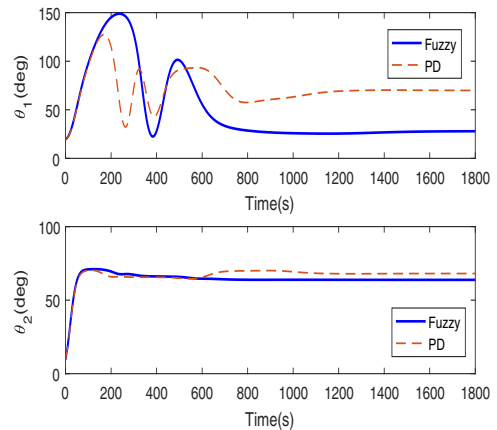


Fig. 8. Time responses for attitude angle about z-axis of the pendulums

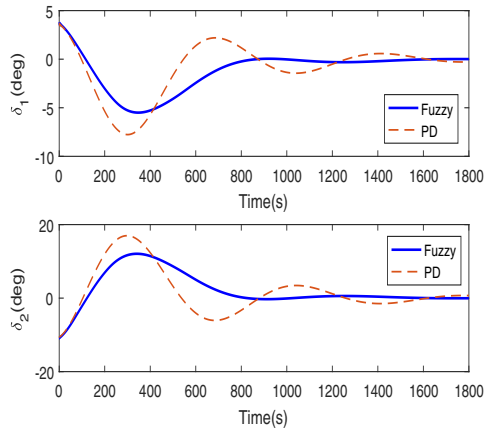


Fig. 9. Time responses for two gimbal deflection angles

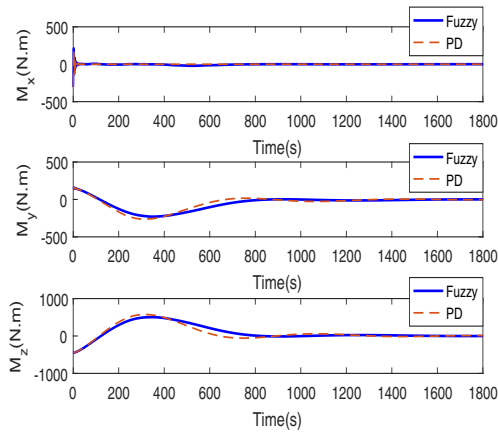


Fig. 10. Time responses for three moments about center of mass of the spacecraft

stable attitude of the spacecraft and fuel sloshing almost like without uncertainties condition. For example, time responses of attitude angle of the pendulums about x-axis and gimbal deflection angles are shown in Figure 11 and Figure 12.

CONCLUSION

We have developed a nonlinear model to describe coupled dynamics of the spacecraft that maneuver in space, and fuel sloshing. A classic and fuzzy controller have been designed to control translation velocity vectors and attitude of the spacecraft, and attitude of the pendulums. A physical example has been included to describe differences between classic and fuzzy controller on designed nonlinear dynamic system. Its shown fuzzy controller is more robust than classic controller when uncertainties are included.

REFERENCES

- [1] M. Navabi and A. Davoodi, "3D multi-pendulum model of slosh dynamics in a spacecraft," *Proceeding of RAST*, 2017, pp. 259–262.
- [2] M. Navabi and M. Radaei, "Attitude adaptive control of space systems," *Proceeding of RAST*, 2013, pp. 973–977.

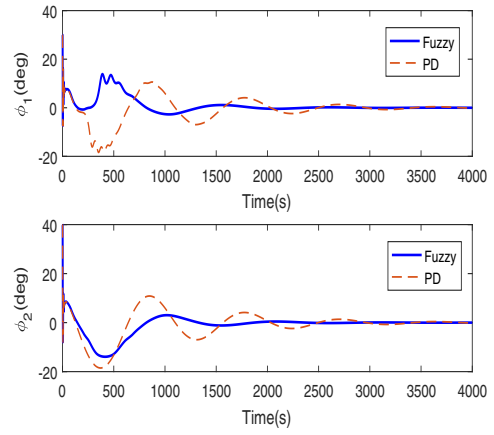


Fig. 11. Time responses for attitude angle of the pendulums about x-axis when uncertainties are included

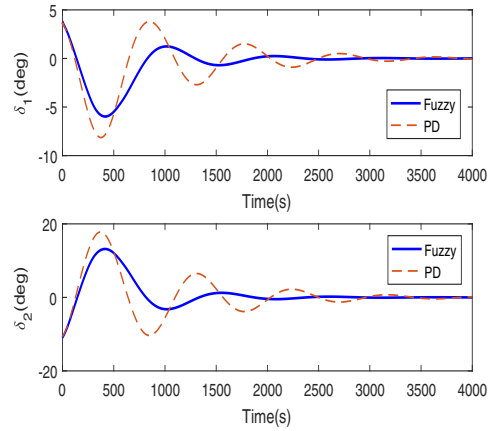


Fig. 12. Time responses for gimbal deflection angles when uncertainties are included

- [3] M. Navabi and H. Rangraz, "Comparing optimum operation of pulse widthpulse frequency and pseudo-rate modulators in spacecraft attitude control subsystem employing thruster," *Proceeding of RAST*, 2013, pp. 625–630.
- [4] M. Reyhanoglu and J. Rubio Hervas, "Nonlinear control of a spacecraft with multiple fuel slosh modes," *Proceeding of CDC-ECC*, 2011, pp. 6192–6197.
- [5] J. Rubio Hervas, M. Reyhanoglu and H. Tang, "Thrust-vector control of a tree-axis stabilized spacecraft with fuel slosh dynamics," *Proceeding of ICCAS*, 2013, pp. 761–766.
- [6] Z. Zhao, M. Tomizuka and S. Isaka, "Fuzzy gain scheduling of PID controllers," *IEEE Trans. Syst., Man, Cybern.*, vol. 23, issue 5, pp. 1392–1398, 1993.